Audit Synthesis: Emirp Dynamics to Visual Grammar, Critical CA, and Feynman Path Integrals

Comprehensive summary of the interdisciplinary investigation as of June 13, 2026. Key threads include emirp number theory, category-theoretic interpretation of AI generative outputs, Cascade CA criticality experiments, topological analysis, and connections to quantum path integrals.

1. Emirp Prime Audits

Residue distributions mod 37 and others showed near-uniformity with mild dynamic correlations. DR transition matrix exhibited statistically significant but modest memory (chi2 approx 239, p << 0.001). Spectral gap large (~0.928). Gap statistics: 11,241 emirps, 397 twins, leptokurtic distribution. Stratification by digit length and reversal graph (mostly 2-cycles) confirmed.

2. Category-Theoretic Framework for AI Visual Grammar

Syntactic category S (visual primitives: branches, closures, symmetries) mapped functorially to semantic M (algebraic structures). Natural realizations constrained by topology and geometry. Persistent homology provides invariants for discretization.

3. Cascade CA Criticality Experiments

Phase sweep identified resonant strength approx 0.52 maximizing Branching Index with emergent structures. Directed Percolation universality class. Persistent homology: non-trivial beta_0 (components) and beta_1 (loops). Critical grid shows branching interference patterns.

4. Feynman Path Integrals on Lattices and Discrete Schrodinger

CA models lattice path sums under potential (mediator_strength). Links to Poisson summation, theta functions, AGM/elliptic integrals. Supports visual physics emergence from local rules.

Conclusion and Next Steps

Framework demonstrates information creation at criticality and functorial mappings. Open: shape grammar extraction, explicit path integral simulations, higher-resolution hybrids.